PARIGALA PRASHANTH KUMAR

AI ENGINEER (FRESHER)

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PROFESSIONAL SUMMARY

AI Engineering student pursuing B.Tech in CSE (AI & ML) with hands-on, self-directed experience building generative AI tools. Followed a structured 26-week roadmap covering LLM APIs, RAG systems, and agentic frameworks (LangChain, LlamaIndex, LangGraph, CrewAI), independently shipped a full-stack local RAG chatbot and a progress-tracking web app, and reverse-engineered a production AI curriculum website end-to-end into a documented architecture teardown and working clone. Comfortable with prompt engineering, rapid prototyping (vibe coding), and learning new AI tooling through hands-on exploration.

TECHNICAL SKILLS

- **Languages & Web:** Python, SQL, HTML, CSS, FastAPI
- **GenAI & LLM Tooling:** LLM APIs, Prompt Engineering, Retrieval-Augmented Generation (RAG), LangChain, LlamaIndex, LangGraph, CrewAI, Fine-tuning, Agentic AI, Ollama (Llama 3.2), ChromaDB / Vector Databases, Embeddings (nomic-embed-text)
- **Tools & Practices:** Vibe Coding / AI-assisted development, UI/UX Design, Git-based workflow, localStorage-based client persistence, Data export/import (CSV/JSON/Markdown), Frontend architecture analysis (vanilla JS, static-site teardowns), Model Context Protocol (MCP), llms.txt

PROJECTS

CONSENSUS — AI Chatbot with RAG System (Local Document Q&A) — 2026

Full-Stack + AI: Frontend, Backend, and RAG Pipeline | Python, FastAPI, HTML/CSS/JS, Ollama, ChromaDB, Llama 3.2

- Built end-to-end: Frontend (CONSENSUS chat UI in HTML/CSS/JS), Backend (FastAPI serving UI + APIs from one host), and AI features (full RAG pipeline).
- RAG in simple words: Upload your PDFs / TXT / MD / DOCX notes, ask questions in plain English, and the chatbot answers only from your documents and shows which file the answer came from — instead of guessing.
- Implemented document indexing: file upload → clean text → split into 500-char chunks → embeddings with nomic-embed-text → store in ChromaDB vector database.
- Implemented Q&A flow: question → retrieve top-3 relevant chunks → Llama 3.2 generates grounded answer with source filenames and anti-hallucination prompt.
- Added upload API, paste-text-as-source, sources list, and health check; runs 100% locally with Ollama for private documents with no cloud API cost.

GenAI Learning Tracker — Personal Progress Dashboard — 2026

Standalone HTML/JS Web App

- Designed and built fully standalone `GenAI_Tracker.html` to track a self-authored 26-week, 182-task Generative AI roadmap.
- Implemented LeetCode-style contribution heatmap with month-by-month color differentiation and visible month gaps.
- Built persistent, dependency-free data layer using browser localStorage for portability.
- Engineered CSV, JSON, and Markdown export with JSON import for cross-device backup and restore.
- Derived all statistics (streaks, completion %, activity logs) live from logged activity; added checklist, activity log, reset-with-confirmation, and real-time Saved indicator.

AI Engineering from Scratch — Full Site Reverse-Engineering & Static Clone — Nov 2026

Frontend Architecture Study | HTML, CSS, Vanilla JavaScript, Vercel, MCP, llms.txt | GitHub

- Reverse-engineered the production site behind a 542-lesson AI engineering curriculum (aiengineeringfromscratch.com) at the browser level and documented its full architecture: static HTML + vanilla-JS frontend, Vercel serverless renderers for lessons/certifications, curriculum-specific Markdown-to-JS build pipeline, and an agent-facing layer (llms.txt, OpenAPI 3.1, MCP server).

- Rebuilt the entire landing page from scratch as a working static clone carrying the full curriculum snapshot (542 lessons, 20 phases, 13 learning paths, 250 glossary terms, 526 reusable artifacts) with zero frameworks and no build step.
- Preserved and re-implemented all real interactions: three-mode theme system, ⌘K command palette with a local search index, per-lesson progress tracking in localStorage, Web Speech narration, autoplaying accessible carousel, i18n picker, and copy feedback.
- Verified the clone against the original with 10 automated interaction checks and side-by-side screenshots (light/dark, mobile, modal, palette); documented deliberate divergences (no analytics, attribution-safe metadata) and published the full teardown (REPORT.md) and code on GitHub.

EDUCATION

B.Tech, Computer Science & Engineering — AI & Machine Learning | Jul 2024 – Jun 2027

Dr. K.V. Subba Reddy Institute of Technology | CGPA: 7.20

Diploma, Computer Engineering | Oct 2021 – Jun 2024

Brindavan Institute of Technology and Science | CGPA: 7.20

SELF-DIRECTED LEARNING

- Progressing through structured 26-week / 182-task Generative AI Engineering roadmap: Python, LLM APIs, LangChain, LlamaIndex, LangGraph, CrewAI, RAG design, deployment.
- Targeting job-ready AI engineering proficiency within six months, tracked via custom tracker app.

LANGUAGES

English (Intermediate) • Telugu (Native)

AREAS OF INTEREST

Agentic AI • Fine-tuning • Retrieval-Augmented Generation (RAG) • Large Language Models (LLMs)